

Total No. of Questions : 8]

SEAT No. :

PD4109

[Total No. of Pages : 3

[6402]-69

S.E. (Artificial Intelligence and Machine Learning)

OPERATING SYSTEMS

(2019 Pattern) (Semester - IV) (218552)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answers Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Define with example Mutual exclusion, race condition, Semaphore, and Deadlock. [8]

b) How is the TestandSet instruction used to solve the critical section problem? Write and explain the pseudocode for TestandSet instruction implementation. [9]

OR

Q2) a) Explain the wait-for graph method for deadlock detection along with an example. [8]

b) For the given snapshot of the system: [9]

	Allocation				Max				Available			
P1	0	1	1	0	5	4	4	4	1	2	4	3
P2	2	0	0	1	2	1	3	3				
P3	1	1	3	2	8	5	4	4				
P4	2	1	2	2	6	5	4	5				
P5	3	1	0	1	6	2	2	3				

Determine:

- i) Need Matrix
- ii) Current snapshot is safe or unsafe.
- iii) Whether following requests can be granted or not for Process P3 : 1 1 0 1

P.T.O.

- Q3) a)** Consider a logical address space of 128 pages of 512 Bytes each mapped onto a physical memory of 32 frames of 512 Bytes. [9]
- i) How many bits are in the physical address?
 - ii) How many bits are in the logical address?
 - iii) How many bits are required to determine the frame number in the physical address?
 - iv) How many bits are required to determine the page number in logical address?
- b) With the help of neat diagrams (Bar diagram and Tree), Write a short note on the Buddy system. [9]

OR

- Q4) a)** Consider the page reference string with 3 frames. [9]
1, 2, 3, 4, 5, 4, 5, 3, 6, 7, 8, 9, 5, 4, 7, 8, 9, 5, 4, 2.
Calculate the number of page faults for the following page replacement algorithm.
- i) FIFO
 - ii) OPTIMAL
 - iii) LRU
- b) For the system where memory cycle time is 200 nanoseconds, Translation Look-aside Buffer access time is 20 nanoseconds, [9]
Determine the Effective Access Time when
- i) Hit ratio is 99%
 - ii) Hit ratio is 95%
- Also justify if the value of hit ratio should be smaller or larger.

- Q5) a)** A disk drive has 200 tracks, numbered 0-199. The drive is currently serving the request at track no 53. The queue of pending requests in FIFO order is 98, 183, 37, 122, 14, 124, 65, 67. Starting from the current head position what is the total distance that disk arm moves to satisfy all the pending requests for the following disk scheduling algorithms. Assume that the head is moving in the increasing order of track number for C-SCAN and LOOK. [12]
- i) FCFS
 - ii) C-SCAN
 - iii) LOOK
 - iv) SSTF
- b) Explain with diagrams different Record Blocking techniques. [6]

OR

- Q6)** a) Explain with the help of neat diagrams different secondary storage management methods. [9]
b) List and explain (any 3) directory structure organization. [9]

- Q7)** a) Explain ORIGIN, EQU, and LTORG with an example. [6]
b) Explain the data structures required for TWO PASS Assembler in detail. [6]
c) Differentiate between literal and immediate operand. [5]

OR

- Q8)** a) What are the types of loaders? Discuss 4 different functions of the loader. [6]
b) Using a suitable diagram, explain the 'Compile and Go' Loader scheme with its advantages and disadvantages. [6]
c) Explain any two phases of the compiler with a suitable diagram. [5]

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